

REPO: spf13/cobra

GitHub: <https://github.com/spf13/cobra>

Type: Go CLI Framework Library

Runtime: Go 1.21+

1. What It Is

Cobra adalah Go library paling populer untuk membangun command-line applications, digunakan oleh kubect!, GitHub CLI (gh), Hugo, dan Terraform. It provides a structured pattern for defining commands, subcommands, flags, and help text. Think of it as the skeleton that turns a collection of Go functions into a polished, documented CLI tool with auto-complete and `--help` generation.

2. Tech Stack

- **Language:** Go 1.21+
 - **Dependencies:** github.com/spf13/pflag (POSIX flag parsing), github.com/spf13/viper (optional config management)
 - **Runtime:** Compiles to single static binary — no runtime dependency on target machine
 - **Companion tool:** `cobra-cli` scaffolding generator
-

3. Kegunaan untuk ResistanceZero

NOW:

- Bangun rz CLI yang menjadi unified entry point untuk semua Python tools di rz-work:
- `rz build pln` — run `tools/build-osm-dataset.py`
- `rz minify` — run `terser + cleancss`
- `rz serve` — start `python3 -m http.server 8081`
- `rz deploy` — push ke GitHub Pages
- Lebih ergonomis dari raw shell scripts karena ada `--help`, flag validation, dan error messages

FUTURE:

- `rz article new --title "FF-4" --series future-forward` — scaffold HTML article dari template
 - `rz check links` — crawl semua HTML files dan report broken links
 - `rz calc validate` — run linting/validation terhadap semua calculator pages
 - Distribute single binary ke kolaborator tanpa perlu install Python environment
-

4. Installation

```
# Install Go first (if not installed)
sudo apt-get install -y golang-go

# Install cobra-cli scaffolding tool
go install github.com/spf13/cobra-cli@latest
```

```
# Scaffold new CLI project
mkdir ~/rz-work/tools/rz-cli && cd ~/rz-work/tools/rz-cli
go mod init github.com/baguspermana7/rz-cli
cobra-cli init

# Add subcommands
cobra-cli add build
cobra-cli add serve
cobra-cli add deploy
```

5. Integrasi

Tool rz-work	Cara Integrasi
Python tools (`build-osm-dataset.py`)	`cmd.RunE` calls `exec.Command("python3", "tools/build-osm-dataset.py")`
Taskfile (go-task)	Cobra dan Task bisa dipakai bersama – Task untuk dev orchestration, Cobra untuk distributable binary
`process-images.py`	`rz images process --input assets/` wraps Python script dengan path validation
GitHub Pages deploy	`rz deploy` wraps `git push origin main` dengan pre-deploy checks

6. Contoh Penggunaan

```
// cmd/serve.go – rz serve command
var serveCmd = &cobra.Command{
    Use: "serve",
    Short: "Start local HTTP server at port 8081",
    RunE: func(cmd *cobra.Command, args []string) error {
        port, _ := cmd.Flags().GetString("port")
        return exec.Command("python3", "-m", "http.server", port).Run()
    },
}

func init() {
    rootCmd.AddCommand(serveCmd)
    serveCmd.Flags().StringP("port", "p", "8081", "Port to serve on")
}

# Build and use the CLI
cd ~/rz-work/tools/rz-cli
go build -o ~/.local/bin/rz .

# Usage
rz serve --port 8081
rz build pln --verbose
rz deploy --dry-run
```

7. Batasan & Risiko

- **Overkill untuk satu orang:** Jika rz-work hanya dikelola sendiri, Taskfile (go-task) sudah cukup untuk orchestration. Cobra worth it hanya jika ada rencana mendistribusikan binary ke orang lain atau ingin CLI yang

sangat polished.

- **Maintenance overhead:** CLI Go perlu di-compile ulang setiap ada perubahan logic. Python scripts lebih cepat untuk iterasi.
- **Tidak menggantikan Python tools:** Cobra adalah wrapper/orchestrator — tidak menggantikan `build-osm-dataset.py` atau `process-images.py`. Tetap butuh Python environment.
- **Binary size:** Compiled Go binary ~5–10 MB untuk project kecil. Tidak cocok jika ingin distribusi via GitHub Pages (file statis).